STA258: Statistics with Applied Probability

Introduction

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Outline

Basics

Definition (Statistics)

Statistics is the science of collecting, classifying, summarizing, analyzing and interpreting data.

Definition (Descriptive Statistics)

Numerical and graphical methods used to analyze, interpret and represent data.

also associate conclusion with confidence

Definition (Inferential Statistics)

Use information from a sample to make generalizations about a population.

calculated and known　　　usually unknown

Outline

Types of Data

Definition (Quantitative data)

Numerical data that can be measured.

we can perform meaningful mathematical calculations with quantitative data

- Examples:
  - Number of hours you studied this week
  - Distance from your house to the university
  - Area (in ft^2) of the floor of a concourse
- Quantitative data can also be divided into:
  - Discrete data → data takes specific values
  - Continuous data ↘ data can take any value within a domain

Types of Data Ctd...

Definition (Discrete data)

Measurements can only take specific vales.

- Examples:
  - The number of heads you get when you toss a coin 5 times
  - The number of rooms in a residence

Types of Data Ctd...

Definition (Continuous data)

Measurements can take any value within a specific ~~range~~ domain of values

- Examples:
  - Height
  - Weight
  - Temperature
  - The time taken to complete a task

Types of Data Ctd...

Temp

− 0 +

Note

Note that quantitative data can also be categorized as:

- *Interval data*
- *Ratio data* (meaningful zero value)

However we will not go into this much detail in this course.

week 2: 5 hrs
week 2: 15 hrs

$\frac{15}{5} = 3$

Temperature
container 1: 50°C
container 2: 100°C Not double

Types of Data Ctd...

Definition (Qualitative data)

Data can not be measured on a numerical scale. Instead the data falls into categories.

- Examples:
  - Favourite flavour of ice-cream
  - Day of the week (Mon, Tue, . . .) that an event occurred
  - City you live in

place objects into bins

often interested in counts

Types of Data Ctd...

Note

Note that qualitative data can also be categorized as:

- *Nominal data* most basic type
- *Ordinal data* categorical data which has a natural ordering

However we will not go into this much detail in this course.

eg Surveys

How much did you like the course

1 highly unsatisfied 2 3 4 5 highly satisfied

Exercise

dress size   0   00
qualitative

	Quantitative data	Qualitative data
Age	✓	
Age cohort <19, 19-25, 26-30, ...		✓
Weight	✓	
Gender		✓
Time taken to run a lap	✓	
Cost of a textbook	✓	
Major at university		✓
Classroom seating capacity	✓	
Interest rate	✓	
Average annual return of a stock	✓	
Shoe size ... 7, 7.5, 8, 8.5, 9, 9.5, ...		✓
Yr of university (Based on # of credits completed)		✓

1, 2, 3, 4

Outline

Types of Studies

potential workaround ethics without conducting observational study: simulations and modelling

Definition (Observational Studies)

We observe units and take measurements without assigning treatments

- Examples: disease outbreaks
  - The effect of smoking on lung capacity.
  - The effect of heroin on brain function.

often performed for ethical reasons or due to inability to control a situation

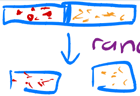
Types of Studies Ctd...

Have a lot of control

Definition (Experimental Studies)

Treatments are assigned to units and then the effects are observed and measured

double blind



experimenter
randomly assigned participants

- Examples:
  - Testing whether the packaging of a product is appealing to consumers before sending it out to the market.
  - Providing a group with windows pc's and another similar group with mac's and measuring the time taken to complete certain tasks.
  - Clinical trials (testing a new pharmaceutical drug or vaccine or treatment etc.)

Types of Studies Ctd...

Definition (Sample surveys)

Data is obtained from a selected part of the population using a survey instrument.

- Similar to an experimental designed study.
- Examples:
  - Satisfaction survey of guests at a resort.
  - Orientation week satisfaction survey.

Types of Studies Ctd...

- Other types of studies include:
  - Case control studies
  - Cohort studies
  - Cross sectional : examining a snapshot of something at a moment in time
  - Prospective studies
  - Retrospective studies
- These will not be discussed but are mentioned for completeness

Outline

Important

Introduction to Inferential Statistics

Definition (Unit)

Objects we are interested in and which measurements are recorded.

Definition (Population)

A (large) group of units that we are interested in studying.

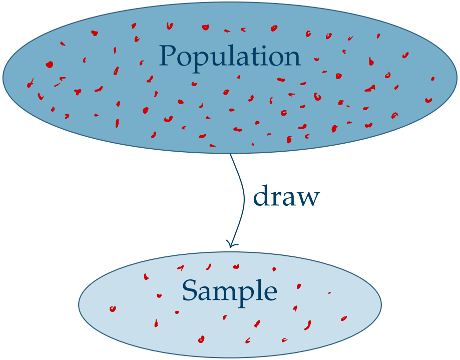
Definition (Sample)

A subset of the population.

Intro to Inferential Statistics Ctd...

Our picture so far:

typically large (relative)



caution: parameter has 2 meanings

Intro to Inferential Statistics Ctd...

Definition (Parameter) (of a population)

A numerical measure of a population.

- Usually not known since it is very hard to take measurements on every unit in a population.
- Parameters of interest for us:
  - μ : Population mean
  - σ : Population standard deviation
  - p : Population proportion

Other potential meaning of parameter in context of statistics

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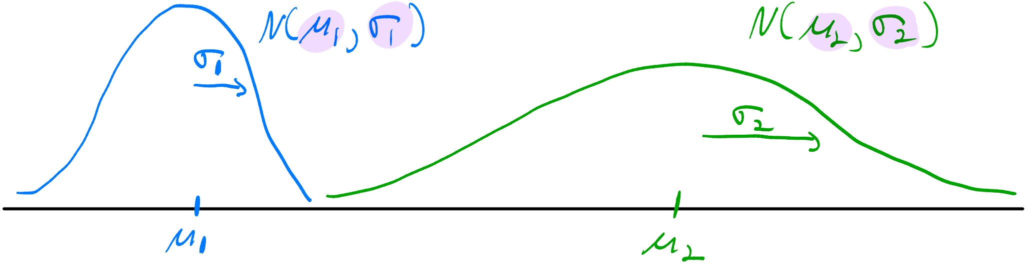
$\downarrow$

Distribution

Numerical values of a distribution which describe features such as centrality, dispersion, Skewness, etc.

Example: Normal Distribution. Parameters: μ, σ



Be a little cautious with use of parameter in context.

Intro to Inferential Statistics Ctd...

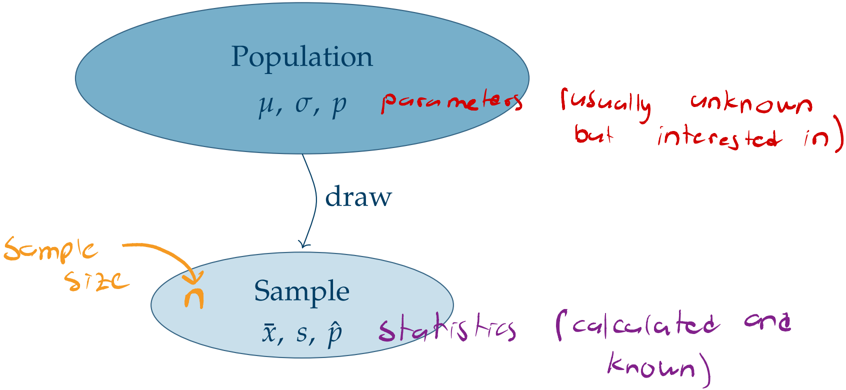
Definition (Statistic)

A numerical measure of a sample.

- Known as it is much easier to take measurements on a sample and calculate statistics.
- Statistics of interest for us:
  - $\bar{x}$: Sample mean
  - s : Sample standard deviation
  - $\hat{p}$: Sample proportion
- A statistic is also often referred to as an estimator.

Intro to Inferential Statistics Ctd...

Updated picture:



Intro to Inferential Statistics Ctd...

Goal (Statistical Inference) (Big part of course)

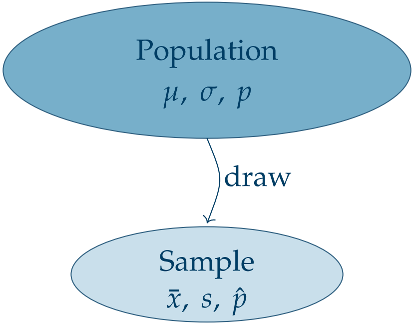
To calculate estimators of population parameters, and to quantify the accuracy of these estimators with probabilities.

- Interested in parameters in a population.
- Calculate statistic from a sample.

$$\underset{\text{known}}{\text{Statistics}} \xrightarrow{\text{Estimate}} \underset{\text{unknown}}{\text{Parameters}}$$

- Quantify how confident we are about a statistic estimating its corresponding parameter.

Intro to Inferential Statistics Ctd...



$$\bar{x} \xrightarrow{\text{Estimate}} \mu$$

$$s \xrightarrow{\text{Estimate}} \sigma$$

$$\hat{p} \xrightarrow{\text{Estimate}} p$$

Outline

Information on Sample Data

- It is usually very difficult (or impossible) to measure every unit in a population.
- It is more feasible and practical to draw a sample and take measurements on all units in this sample.
- We would like our sample to be representative of the population it is drawn from.
- This is because we would like to make generalizations about the population based on our sample.

$$\text{Sample} \xrightarrow{\text{Generalize}} \text{Population}$$

Types of samples

N

$\frac{1}{N}$

Definition (Random Sample)

Each unit in a population has an equal chance of being selected. Also each of combination of size n has an equal chance of selection.

- Other sampling techniques include:
  - Stratified sampling
  - Cluster sampling
  - Multistage sampling

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- However these will not be covered in this course

Issues with Collecting Sample data

- Collecting sample data is expensive (in terms of time and money).
- Data may be biased:

Selection bias	:	Sample is not representative of the population since a subset of the population has no chance of being selected for the sample
Nonresponse bias	:	There may be a reason that certain respondents refuse to participate. As such we lose information.
Measurement error bias	:	The response measured and recorded for an individual unit is not correct.

- Ideally we would like to draw a random sample.